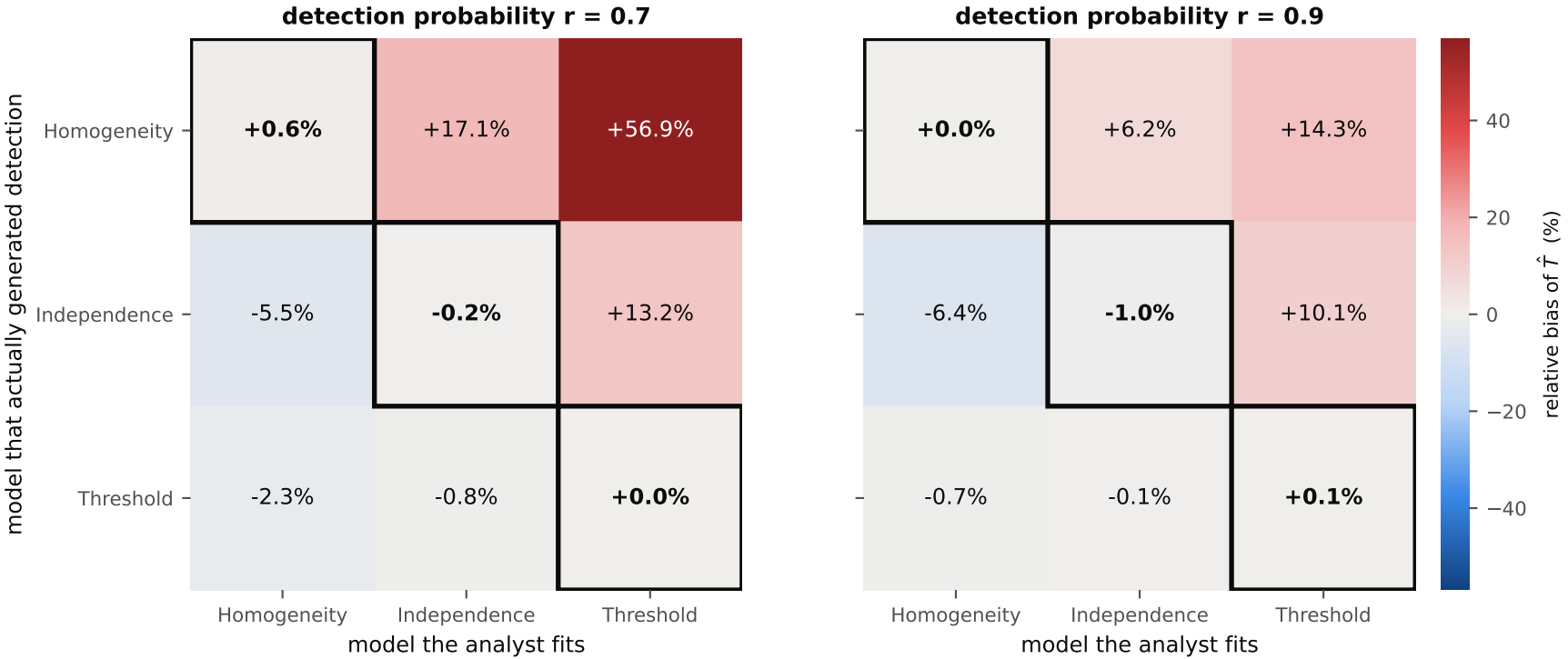
Choosing the detection model matters more than sampling error
detection probability r = 0.7
detection probability r = 0.9
model that actually generated detection
Homogeneity
Independence
Threshold
+0.6%
+17.1%
+56.9%
-5.5%
-0.2%
+13.2%
-2.3%
-0.8%
+0.0%
+0.0%
+6.2%
+14.3%
-6.4%
-1.0%
+10.1%
-0.7%
-0.1%
+0.1%
Homogeneity
Independence
Threshold
model the analyst fits
model the analyst fits
relative bias of $\hat{T}$ (%)
40
20
0
−20
−40
Boxed cells are correctly specified. 1,000 replicates per cell; 100 active collars, 30 groups, random mixing.